

Model A · Diabetic Retinopathy Grading · Full Fine-tune

All RETFound-DINOv2 layers updated · Layer-wise LR decay · 50 epochs · 4-class (R0/R1/R2/R3A)

Upgrade from Linear Probe → Full Fine-tune

LP kept the ~307 M backbone FROZEN — only a single linear head was trained, measuring how much structure was already encoded.

FT UNFREEZES all layers with layer-wise LR decay (factor 0.65):
deepest blocks get full LR; shallowest get ~0.0002× LR.
This allows non-linear class boundaries that a linear head cannot form.

0.915

Test AUROC (LP 0.821)

0.624

Macro Sensitivity (LP 0.561)

0.913

Macro Specificity (LP 0.865)

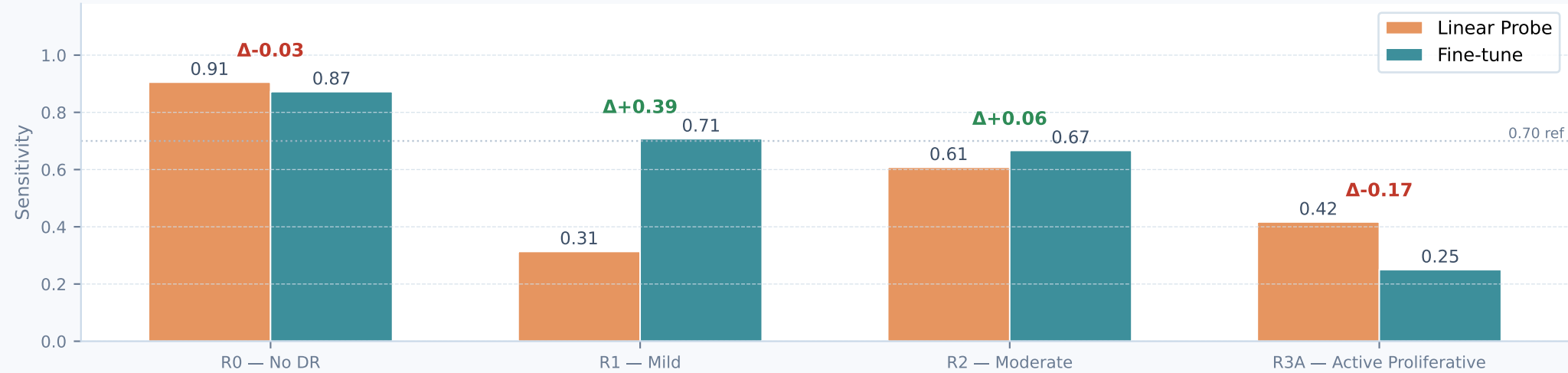
FT Settings (differences from Linear Probe)

Backbone	FULLY UNFROZEN (LP: frozen — no gradient)
LR scheme	Layer-wise decay ×0.65 per depth level
Batch size	24 (LP: 64 — reduced to fit full backprop GP)
Best epoch	19 (LP: 38 — early convergence with unfroz)
Epochs trained	50 · Best ckpt saved by val AUROC
Loss / weights	CrossEntropyLoss · R0=1.00 R1=1.79 R2=9.
Test eval	Best ckpt reloaded → single pass on held-out

LP vs FT · Test Set Metrics

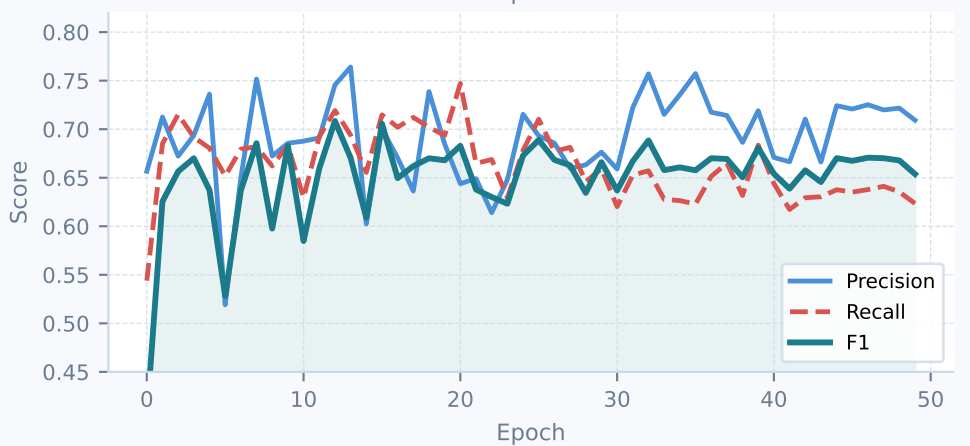
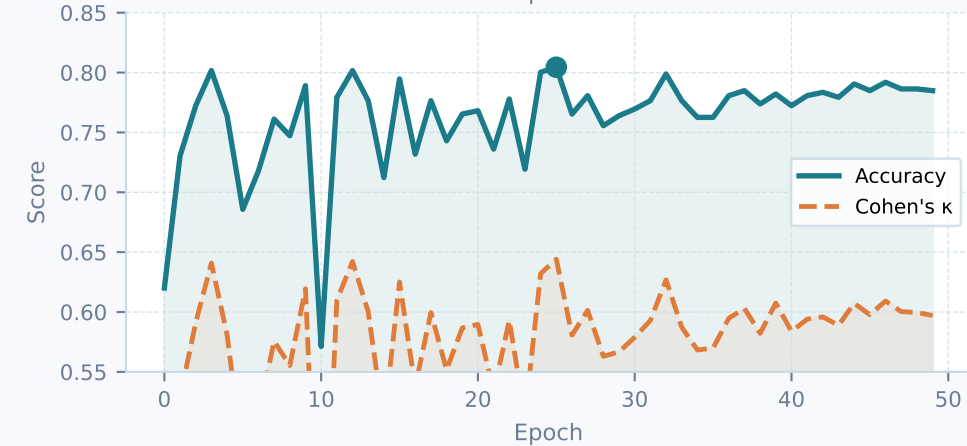
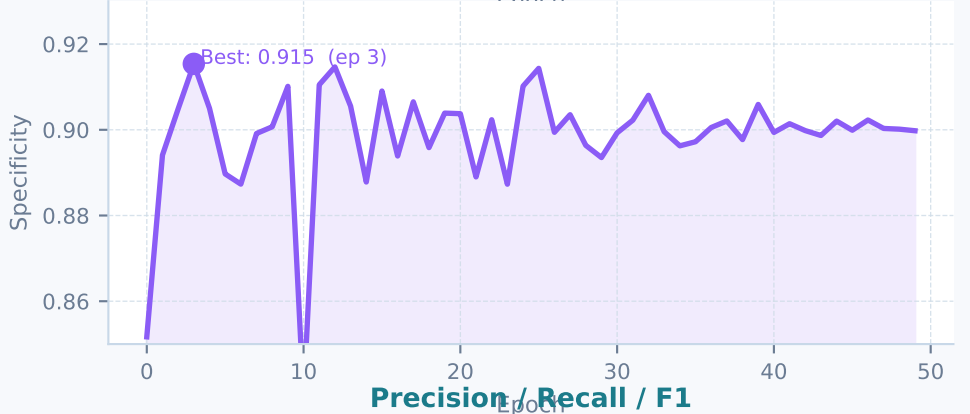
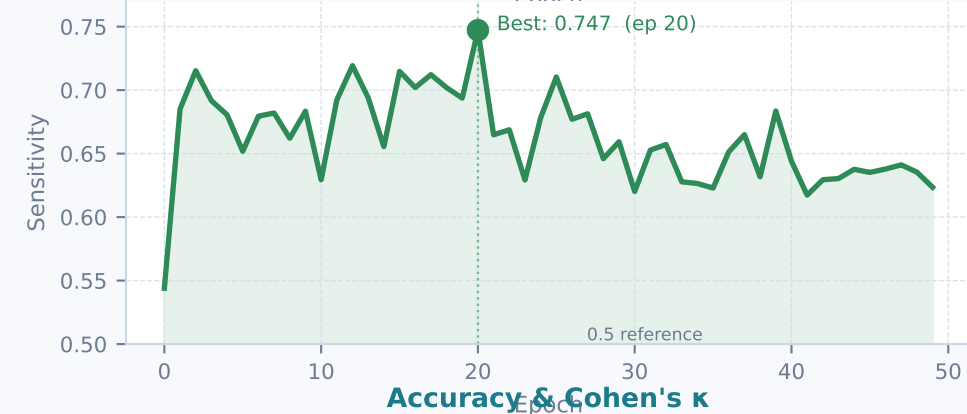
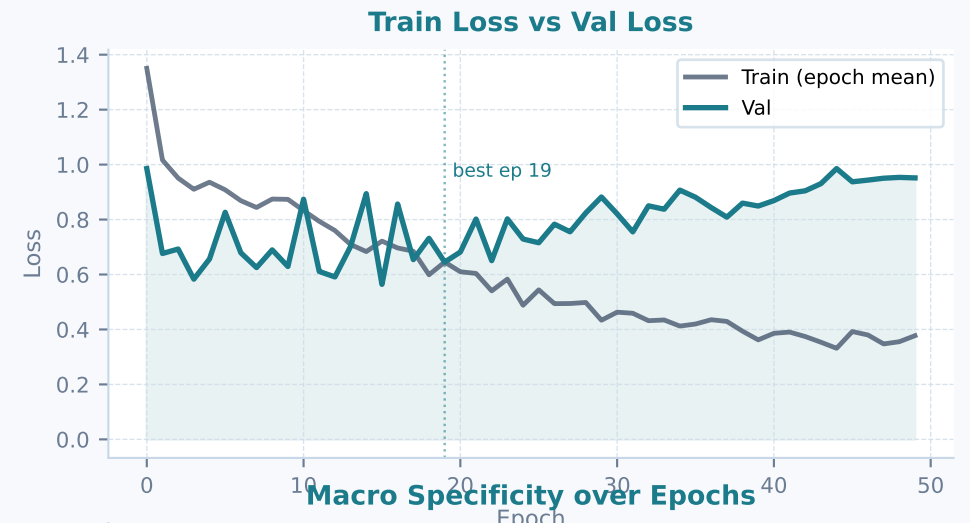
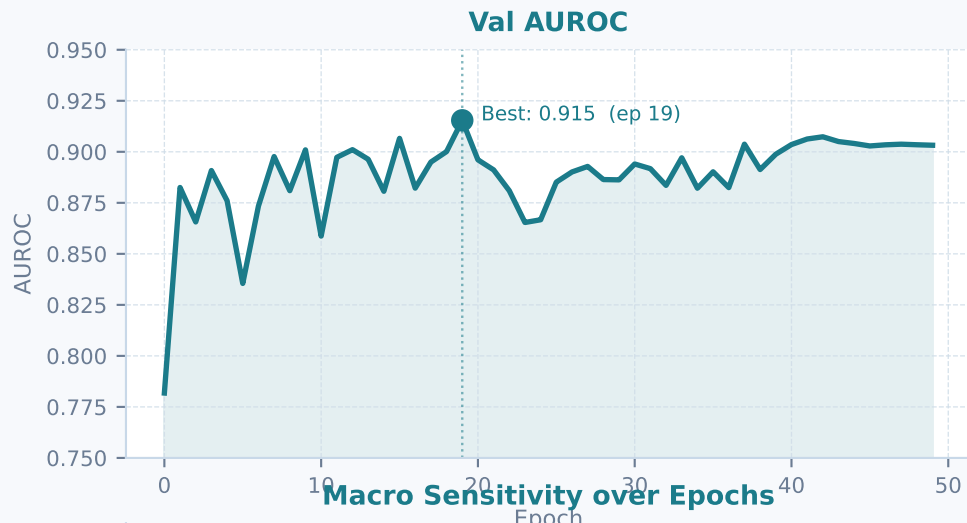
Metric	Linear Probe	Fine-tune	Δ
AUROC	0.821	0.915	+0.094
Accuracy	66.8%	78.1%	+11.3%
F1 (macro)	0.499	0.624	+0.125
Kappa	0.425	0.618	+0.193
Macro Sensitivity	0.561	0.624	+0.063
Macro Specificity	0.865	0.913	+0.047

Per-class Sensitivity: Linear Probe vs Fine-tune (test set)



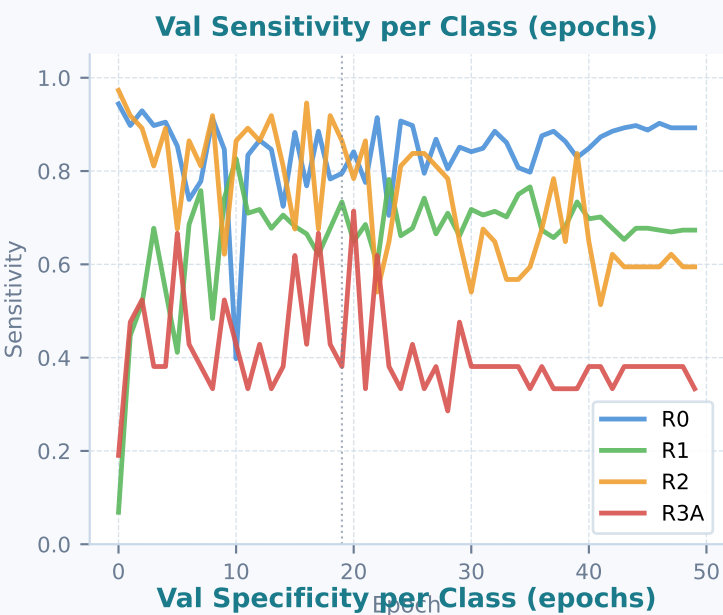
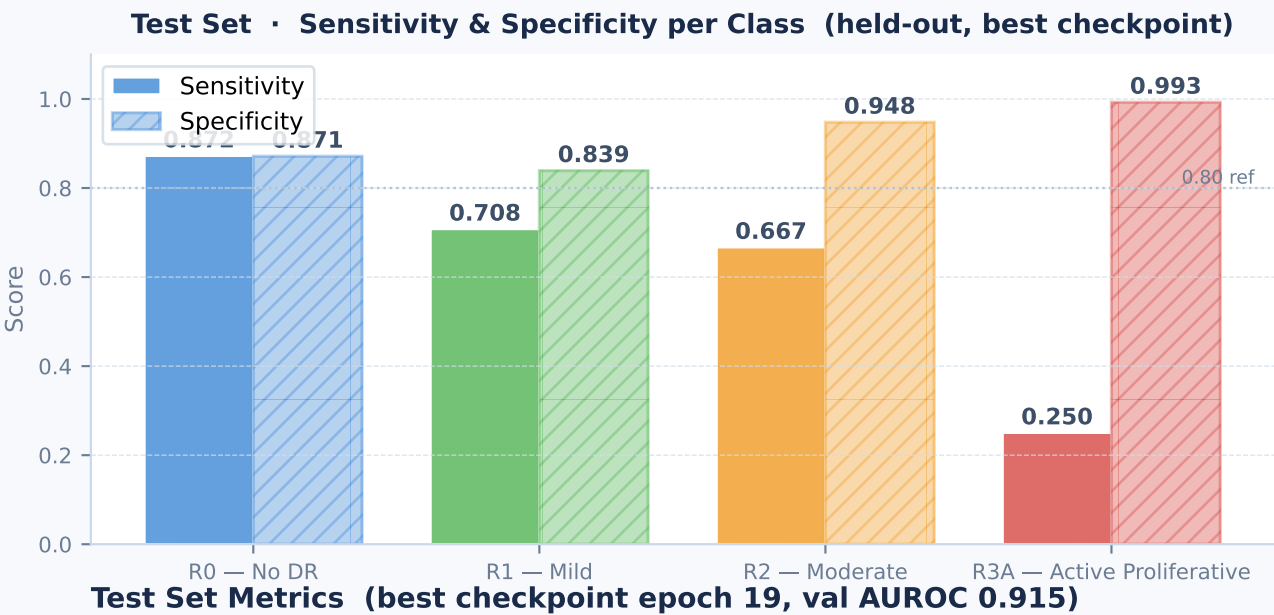
Training Dynamics — All Validation Metrics over 50 Epochs

Best checkpoint at epoch 19 (val AUROC 0.9154) · All backbone layers updated each step



Per-class Sensitivity & Specificity — Full Fine-tune (TEST SET)

Sensitivity = TP/(TP+FN) · Specificity = TN/(TN+FP) · OvR decomposition at best-AUROC checkpoint



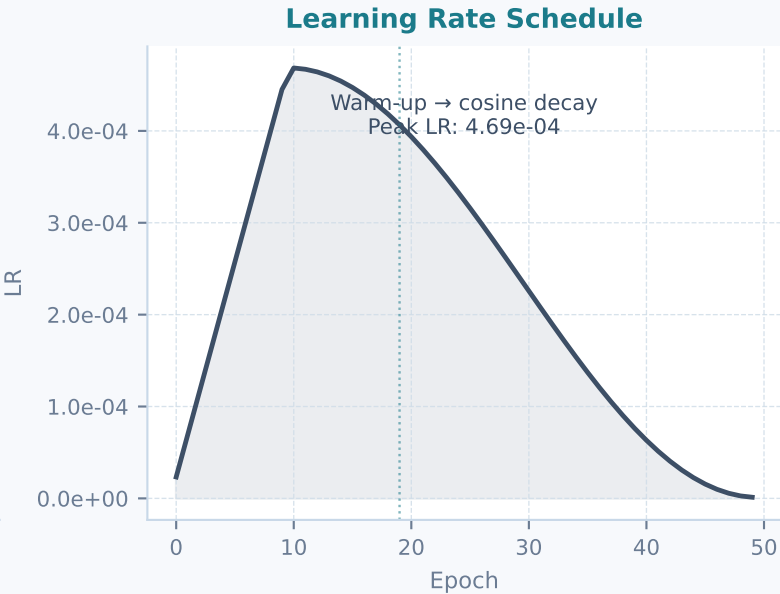
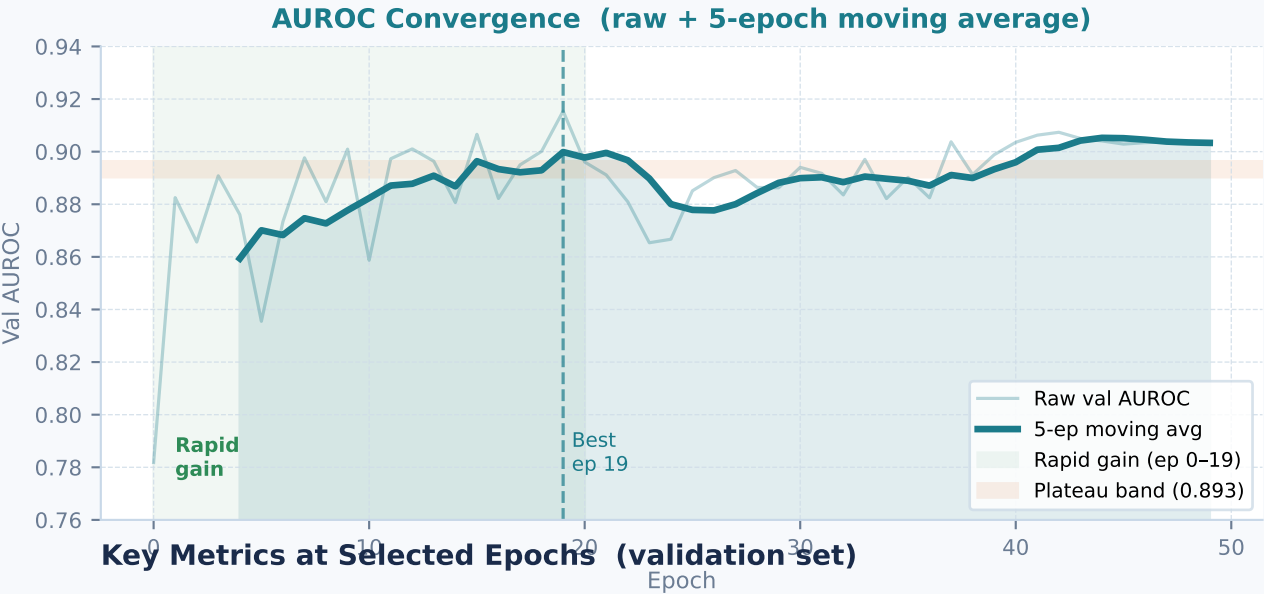
Class	Sensitivity	Specificity	Clinical Note
R0 — No DR	0.872	0.871	Excellent recall; ~15% false alarm rate acceptable for screening
R1 — Mild	0.708	0.839	Major FT gain (+0.394 vs LP); mild DR now well-separated
R2 — Moderate	0.667	0.948	Above LP (0.608); Δ+0.059 — small test n inflates variance
R3A — Active Prolif.	0.250	0.993	Critical gap: 75% of proliferative cases missed on test set (n=48 images)
Macro average	0.624	0.913	Test AUROC 0.915 · Accuracy 78.1% · Kappa 0.618

Clinical Interpretation of Fine-tuned Per-class Performance

- **R3A sensitivity (0.250):** 75% of proliferative cases go undetected — the most dangerous failure. Low test n (~48) means each missed prediction moves sensitivity by ~0.021; interpret with caution.
- **R1 gain (+0.394 vs LP):** Biggest win of fine-tuning: mild DR sensitivity moved from 0.314 to 0.708. Backbone adaptation enables non-linear class boundaries the frozen linear head could not form.
- **R2 gain (+0.059 vs LP):** Moderate DR sensitivity moved from 0.608 (LP) to 0.667 (FT). FT successfully recovers moderate-grade detection.

Convergence Analysis · LR Schedule · Milestone Epochs

Best ckpt at epoch 19 · Plateau ~0.893 from epoch 10 onwards · Val loss divergence = overfitting signal



Epoch	AUROC	Accuracy	Kappa	Macro Sensitivity	Macro Specificity	Train Loss	Val Loss
Ep 0	0.7818	0.6201	0.3456	0.5440	0.8517	1.3504	0.9855
Ep 3	0.8908	0.8017	0.6409	0.6917	0.9154	0.9102	0.5830
Ep 19 ★	0.9154	0.7654	0.5867	0.6937	0.9039	0.6453	0.6452
Ep 12	0.9011	0.8017	0.6423	0.7193	0.9147	0.7601	0.5911
Ep 24	0.8667	0.8003	0.6321	0.6782	0.9102	0.4880	0.7290
Ep 49	0.9032	0.7849	0.5970	0.6235	0.8998	0.3780	0.9516

Key Observations

- AUROC jumped 0.782 → 0.915 in just 19 epochs — pretrained backbone features activate rapidly when all layers receive gradients.
- Val AUROC peaked at epoch 19 then oscillated ~0.893 while val loss climbed 0.65→ 0.95: a clear overfitting signal. Early stopping (patience ≈ 10 on val AUROC) would save ~30 epochs of compute.
- Train loss continued declining to 0.331 by epoch 44, confirming the model kept learning on training data after val AUROC plateaued— the gap between train and val loss widens steadily past epoch 10.
- LR warm-up ramps over the first 10 epochs, peaking at 4.69e-04, then cosine decays to near-zero— this schedule matches the AUROC curve's rapid early rise followed by diminishing returns.
- Sensitivity and specificity at the best checkpoint (epoch 19) reflect the sharpest moment of generalisation; later epochs improve train metrics but not val/test performance.
- Recommended next step: add early stopping to the fine-tune script to avoid unnecessary epochs; then proceed to Model B (maculopathy binary task).